

Multiple Choice (10 marks)

1. The solid-state structures of the principal allotropes of elemental boron are made up of which of the following structural units?
 - (a) B_{12} icosahedra
 - (b) B_8 cubes
 - (c) B_6 octahedra
 - (d) B_4 tetrahedra
2. What is the orbital angular momentum quantum number, l , of the electron that is most easily removed when ground-state aluminum is ionized?
 - (a) 3
 - (b) 2
 - (c) 1
 - (d) 0
3. Infrared (IR) spectroscopy is useful for determining certain aspects of the structure of organic molecules because
 - (a) all molecular bonds absorb IR radiation
 - (b) IR peak intensities are related to molecular mass
 - (c) most organic functional groups absorb in a characteristic region of the IR spectrum
 - (d) each element absorbs at a characteristic wavelength
4. Reduction of D-xylose with $NaBH_4$ yields a product that is a
 - (a) racemic mixture
 - (b) single pure enantiomer
 - (c) mixture of two diastereomers in equal amounts
 - (d) meso compound
5. Which of the following is a primary standard for use in standardizing bases?
 - (a) Ammonium hydroxide
 - (b) Sulfuric acid
 - (c) Acetic acid
 - (d) Potassium hydrogen phthalate

True / False (10 marks)

Answer True or False

6. True or False: At standard temperature and pressure, Br_2 is a red-brown volatile liquid.
7. True or False: BeCl_2 is more likely to act as a Lewis acid than SnCl_2 because it has a higher electronegativity.
8. True or False: Protons and neutrons possess orbital angular momentum only.
9. True or False: Tetramethylsilane is used as a reference because its protons are not strongly shielded in NMR spectroscopy.
10. True or False: The normal modes of a carbon dioxide molecule that are infrared-active include symmetric stretching only.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the factors that contribute to the limiting high-temperature molar heat capacity at constant volume (C_V) of a gas-phase diatomic molecule, including its derivation and implications.
12. Discuss the implications of the Heisenberg uncertainty principle for a quantum mechanical particle in the lowest energy level of a one-dimensional box, focusing on position and momentum.

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